NEXTGEN HACKATHON

VIGIL

Virtual Intelligent Guard for Intrusion and Links

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PS-19

Phishing Website Detection & Real-Time Browser Warning System

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PHISHING THREAT DETECTION PROBLEM

- Phishing is a primary cause of credential theft and financial fraud
- Users often fail to distinguish legitimate and malicious websites existing solutions rely on static blacklists, ineffective against new attacks
- Sensitive information is often compromised before detection
- Excessive alerts in current systems lead to alert fatigue and user negligence

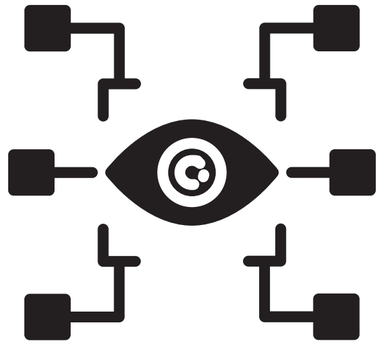
OVER 90% OF CYBERATTACKS START WITH PHISHING, MAKING IT THE MOST COMMON ENTRY POINT FOR SECURITY BREACHES.

OUR SOLUTION ?

V I G I L

- VIGIL is an intelligent cybersecurity system designed to detect phishing threats in real time
- Analyzes URLs, webpage content, and behavioral patterns to identify malicious intent
- Provides instant warnings before users submit sensitive information
- Uses AI-based detection to identify both known and unknown
- (zero-day) attacks
- Ensures low false positives to avoid alert fatigue

WHAT MAKES VIGIL UNIQUE ?



- Unlike traditional systems, VIGIL uses AI-based behavioral analysis instead of static blacklists
- Capable of detecting zero-day phishing attacks (previously unseen threats)
- Provides real-time protection before user interaction, not after damage
- Combines multiple detection layers (URL, content, behavior)
- Designed to minimize false positives, reducing alert fatigue

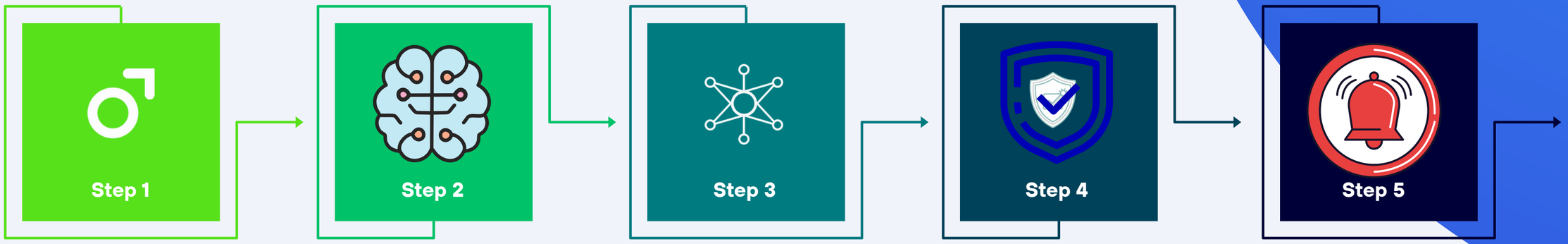
" WATCHFUL PROTECTION "

HOW VIGIL WORKS

- Collects input from URL and webpage content in real time
- Extracts key features such as URL structure, domain patterns, and security indicators
- Processes data through an AI-based detection model
- Classifies the website as safe or phishing based on learned patterns
- Generates instant warning alerts before user interaction

AI-Powered Phishing Detection System

Five-step workflow for intelligent threat identification and prevention



URL Input

Collects website URL and webpage data in real time

Feature Learning

Learns from URL features, domain behavior, and structural anomalies

ML Model Analysis

Uses machine learning to identify phishing patterns

Intelligent Detection

Detects known and zero-day attacks, adapts to evolving threats

Alert

Classifies results with high accuracy and generates warning before user interaction

KEY FEATURES OF VIGIL

- Real-time phishing detection
- AI-based threat analysis
- Pre-warning system before data entry
- Detection of zero-day (unknown) attacks
- Low false positive rate for better user experience

LIGHTWEIGHT AND SCALABLE DESIGN

VIGIL vs Traditional Systems

Traditional Systems	VIGIL
Static, rule-based detection	Intelligent AI-driven detection
Reliance on outdated blacklists	Detects zero-day and evolving threats
Reactive response after attack	Proactive prevention before user action
High false positive rates	Optimized accuracy with reduced false alerts
Limited adaptability to new threats	Continuously learning and adaptive system

FROM REACTIVE DEFENSE TO PROACTIVE PROTECTION.

USE CASES & IMPACT

- Helps users avoid phishing scams and financial fraud
- Enhances safe browsing for students and professionals
- Can be deployed in organizations for security monitoring
- Reduces risk of data breaches and credential theft

We are also working on extending this into a more advanced system-level security tool, integrating deeper behavioral analysis.